

Arunabha Majumder

Mechanical Design Engineer | SolidWorks | Manufacturing Drawings | DfM | Prototyping & Testing

Aalborg, Denmark | +45 71 62 24 00 | somrkmv1997@gmail.com |
linkedin.com/in/arunabha-majumder-681264107

PROFILE

Mechanical Design Engineer with 3 years of hands-on mechanical design experience using SolidWorks, plus a BTech in Mechanical Engineering and an MTech in Mechatronics. Designs and develops mechanical components and assemblies, creates detailed 3D models, manufacturing drawings and engineering documentation, and supports products from concept, prototyping and testing through to manufacture. Applies Design for Manufacture (DfM) and Design for Assembly (DfA) principles, and supports prototype build, assembly and validation hands-on. Works effectively in cross-functional engineering teams with strong problem-solving, communication, attention to detail and organisational skills. Certified SolidWorks Associate (CSWA).

KEY SKILLS

- **Mechanical Design:** Mechanical design, design and development of mechanical components and assemblies, CAD modelling
- **SolidWorks & CAD:** SolidWorks (CSWA certified), detailed 3D models, manufacturing drawings, engineering drawings, technical documentation, Fusion 360
- **Manufacturing:** Design for Manufacture (DfM), Design for Assembly (DfA), machining, additive manufacturing / 3D printing, fabrication and assembly processes
- **Product Development:** Concept design, prototyping, prototype build and assembly, product testing, validation, custom test rigs, Finite Element Analysis (FEA)
- **Teamwork & Delivery:** Cross-functional and multidisciplinary engineering teams, problem-solving, communication, attention to detail, organisation

PROFESSIONAL EXPERIENCE

Research Engineer – Mechanical Design, Robotics & Actuator Development | 2023 – Present *Aalborg University, Aalborg, Denmark*

- Designed and developed mechanical components and assemblies in SolidWorks for an energy-efficient wearable electromechanical product, from concept through prototyping and testing to a working system.
- Created detailed 3D models, manufacturing drawings and engineering documentation used to machine, 3D print and assemble the parts.
- Applied DfM and DfA principles to keep parts practical to manufacture and assemble; reduced cost, mass and part count by eliminating an external force/torque sensor from a robot design.
- Used FEA to iterate designs for strength, stiffness and weight before manufacture, and validated design models against physical testing to below 10% RMS error.
- Supported prototype build, assembly and validation hands-on, integrating motors, encoders, force sensors and data acquisition, and designing custom automated test rigs.
- Improved system efficiency, cutting energy consumption by more than 25% through a new hybrid actuation design (Best Paper Award, IFToMM ISRM 2026).
- Worked with a multidisciplinary team across mechanical, electronics and controls to resolve technical issues, and planned and delivered a 12-participant validation test programme.

MTech Researcher – Mechanical Design & Development, Actuator Systems | 2020 – 2022 *CSIR-Central Mechanical Engineering Research Institute (CMERI), National Laboratory, Durgapur, India*

- Designed, fabricated and assembled pneumatic actuators in-house, and built the test set-up with valves, pressure sensors and data acquisition.
- Tested and characterised actuator performance and improved positioning accuracy to 0.3–0.78% tracking error with a new controller.

BTech Capstone – Mechanical Design Lead, Rehabilitation Device | 2018 – 2019

Siddaganga Institute of Technology, Tumakuru, India

- Led CAD design, manufacture, assembly and user testing of a rehabilitation device; won the Best Major Project Award and an INR 100,000 TATA Technologies Innovation Grant.

EDUCATION

PhD Candidate, Mechatronics and Robotics | 2023 – Present

Aalborg University, Denmark

MTech, Mechatronics | 2020 – 2022

AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), India

BTech, Mechanical Engineering | 2015 – 2019

Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), India

CERTIFICATIONS AND AWARDS

- Certified SolidWorks Associate (CSWA) – Mechanical Design
- Best Paper Award, IFToMM International Symposium on Robotics and Mechatronics (2026)
- TATA Technologies Innovation Award – INR 100,000 Grant (2019)

ENGINEERING PROJECTS

- **Go-Kart (2017):** Designed, fabricated and assembled a custom Go-Kart.
- **Efficycle (2018):** Designed and built a human-electric hybrid vehicle in a multidisciplinary team.

TECHNICAL SKILLS

- **CAD and Simulation:** SolidWorks, Fusion 360, FEA
- **Software:** MATLAB/Simulink, Python, LabVIEW (basic), Arduino
- **Prototyping and Test:** 3D printing (Bambu Lab, Ultimaker Cura), force/torque sensors, data acquisition, test rigs

LANGUAGES

English (C1, fluent) | Bengali (Native) | Hindi (Conversational)

REFERENCES

Available upon request.